

C Program_14.c X

C Program_14.c > ...

```
1  /*PROGRAM TO CHECK WHETHER A GIVEN CHARACTER IS CAPITAL, LETTER, SMALL CASE LETTER, A DIGIT OR A
2  SPECIAL SYMBOL. */
3
4
5  #include <stdio.h>
6
7  int main()
8  {
9      char ch;
10
11      printf("Enter a character: ");
12      scanf("%c", &ch);
13
14      if (ch >= 'A' && ch <= 'Z')
15          printf("Capital Letter");
16
17      else if (ch >= 'a' && ch <= 'z')
18          printf("Small Case Letter");
```

Problems Output Debug Console Terminal Ports

- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_14.c -o Program_14 } ; if (\$?) { .\Program_14 }
Enter a character: P
Capital Letter
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_14.c -o Program_14 } ; if (\$?) { .\Program_14 }
Enter a character: g
Small Case Letter
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_14.c -o Program_14 } ; if (\$?) { .\Program_14 }
Enter a character: 7
Digit
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_14.c -o Program_14 } ; if (\$?) { .\Program_14 }
Enter a character: -
Special Symbol
- PS C:\CCC lab solutions> █

C Program_15.c X

C Program_15.c > main()

```
1 // PROGRAM TO READ MARKS OF FIVE SUBJECT AND PRINT DIVISION.
2
3 #include <stdio.h>
4
5 int main()
6 {
7
8     int a, b, c, d, e, sum;
9
10    printf("Enter you marks out of 100 of");
11
12    printf("\nSubject_1: ");
13    scanf("%d", &a);
14
15    printf("\nSubject_2: ");
16    scanf("%d", &b);
17
18    printf("\nSubject_3: ");
```

Problems Output Debug Console Terminal Ports

```
● PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if ($?) { gcc Program_15.c -o Program_15 } ; if ($?) { .\Program_15 }
Enter you marks out of 100 of
Subject_1: 98

Subject_2: 79

Subject_3: 78

Subject_4: 85

Subject_5: 97
Division B
○ PS C:\CCC lab solutions> 
```

C Program_16.c X

C Program_16.c > main()

```
1 // PROGRAM TO READ THREE NUMBERS AND PRINT MAX.
2
3 #include <stdio.h>
4
5 int main()
6 {
7     int a, b, c;
8
9     printf("\nEnter first number: ");
10    scanf("%d", &a);
11
12    printf("\nEnter second number: ");
13    scanf("%d", &b);
14
15    printf("\nEnter third number: ");
16    scanf("%d", &c);
17
18    if (a > b)
```

Problems Output Debug Console Terminal Ports

● PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_16.c -o Program_16 } ; if (\$?) { .\Program_16 }

Enter first number: 65

Enter second number: 78

Enter third number: 98

98 is maximum

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C Program_17.c X

C Program_17.c > main()

```
1 // PROGRAM TO READ THREE NUMBERS AND PRINT MAX USING LOGICAL OPERATOR.(&&)
2
3 #include <stdio.h>
4
5 int main()
6 {
7     int a, b, c;
8
9     printf("Enter first number: ");
10    scanf("%d", &a);
11
12    printf("Enter second number: ");
13    scanf("%d", &b);
14
15    printf("Enter third number: ");
16    scanf("%d", &c);
17
18    if (a > b && a > c)
```

Problems Output Debug Console Terminal Ports

```
● PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if ($?) { gcc Program_17.c -o Program_17 } ; if ($?) { .\Program_17 }
Enter first number: 56
Enter second number: 87
Enter third number: 39
87 is maximum
○ PS C:\CCC lab solutions> 
```

C Program_18.c X

C Program_18.c > ...

```
1 // PROGRAM TO READ THREE NOS. AND PRINT MAX USING CONDITIONAL (? :) OPERATOR.
2
3 #include <stdio.h>
4
5 int main()
6 {
7     int a, b, c, max;
8
9     printf("Enter three numbers: ");
10    scanf("%d %d %d", &a, &b, &c);
11
12    max = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);
13
14    printf("Maximum number = %d", max);
15
16    return 0;
17 }
18
```

Problems Output Debug Console Terminal Ports

- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_18.c -o Program_18 } ; if (\$?) { .\Program_18 }
Enter three numbers: 45
56
93
Maximum number = 93
- PS C:\CCC lab solutions>

C Program_19.c X

C Program_19.c > ...

```
1  /*ROGRAM TO READ A CHARACTER AND CHECK WHETHER IT IS A SMALL CASE LETTER OR NOT USING
2  CONDATIONAL OPERATOR */
3
4  #include <stdio.h>
5
6  int main()
7  {
8      char ch;
9
10     printf("Enter a character: ");
11     scanf("%c", &ch);
12
13     (ch >= 'a' && ch <= 'z') ? printf("Small Case Letter") : printf("Not a Small Case Letter");
14
15     return 0;
16 }
17
```

Problems Output Debug Console Terminal Ports

- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_19.c -o Program_19 } ; if (\$?) { .\Program_19 }
Enter a character: H
Not a Small Case Letter
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_19.c -o Program_19 } ; if (\$?) { .\Program_19 }
Enter a character: e
Small Case Letter
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_19.c -o Program_19 } ; if (\$?) { .\Program_19 }
Enter a character: P
Not a Small Case Letter
- PS C:\CCC lab solutions> █

C Program_20.c X

C Program_20.c > main()

```
1 //PROGRAM WHICH WILL WORK LIKE A SIMPLE CALCULATOR USING SWITCH-CASE.
2
3 #include <stdio.h>
4
5 int main()
6 {
7     float a, b, result;
8     char op;
9
10    printf("Enter two numbers: ");
11    scanf("%f %f", &a, &b);
12
13    printf("Enter operator (+, -, *, /): ");
14    scanf(" %c", &op);
15
16    switch(op)
17    {
18        case '+':
```

Problems Output Debug Console Terminal Ports

```
● PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if ($?) { gcc Program_20.c -o Program_20 } ; if ($?) { .\Program_20 }
Enter two numbers: 4
5
Enter operator (+, -, *, /): *
Result = 20.00
○ PS C:\CCC lab solutions> █
```

C Program_21.c X

C Program_21.c > main()

```
1  /*PROGRAM TO READ TWO NOS. AND PERFORM SPECIFIC TASK (USING ARITHMETIC OPERATOR)AND THIS
2  PERFORM USING SWITCH-CASE.(CASE IN CHAR FORM). */
3
4  #include <stdio.h>
5
6  int main()
7  {
8      int a, b;
9      char op;
10
11     printf("Enter two numbers: ");
12     scanf("%d %d", &a, &b);
13
14     printf("Enter an operator (+, -, *, /): ");
15     scanf(" %c", &op);
16
17     switch(op)
18     {
```

Problems Output Debug Console Terminal Ports

```
● PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if ($?) { gcc Program_21.c -o Program_21 } ; if ($?) { .\Program_21 }
Enter two numbers: 5
4
Enter an operator (+, -, *, /): *
Multiplication = 20
○ PS C:\CCC lab solutions> 
```

C Program_22.c X

Program_22.c > main()

```
1 // PROGRAM TO CALCULATE FACTORIAL OF A NO.
2
3 #include <stdio.h>
4
5 int main()
6 {
7     int a, n;
8     int num = 1;
9
10    printf("Enter a number:");
11    scanf("%d", &n);
12
13    for (a = 1; a <= n; a++)
14    {
15        num = num * a;
16    }
17
18    printf("The factorial of the given number is: %d", num);
```

Problems Output Debug Console Terminal Ports

- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_22.c -o Program_22 } ; if (\$?) { .\Program_22 }
Enter a number:5
The factorial of the given number is: 120
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_22.c -o Program_22 } ; if (\$?) { .\Program_22 }
Enter a number:8
The factorial of the given number is: 40320
- PS C:\CCC lab solutions> █

C Program_23.c X

C Program_23.c > main()

```
1 //PROGRAM TO READ TWO NOS. AND CALCULATE POWER WITHOUT USING HEADER FILE(<MATH.H>).
2
3 #include <stdio.h>
4
5 int main()
6 {
7     int base, exponent, i, power = 1;
8
9     printf("Enter base: ");
10    scanf("%d", &base);
11
12    printf("Enter exponent: ");
13    scanf("%d", &exponent);
14
15    for(i = 1; i <= exponent; i++)
16    {
17        power = power * base;
18    }
```

Problems Output Debug Console Terminal Ports

- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_23.c -o Program_23 } ; if (\$?) { .\Program_23 }
Enter base: 2
Enter exponent: 4
Power = 16
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_23.c -o Program_23 } ; if (\$?) { .\Program_23 }
Enter base: 7
Enter exponent: 3
Power = 343
- PS C:\CCC lab solutions> █

C Program_24.c X

C Program_24.c > ...

```
1  /*WRITE A C PROGRAM TO FIND THE VALUE OF Y USING
2
3      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
4      1+x/n      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
5  Y(x,n) = 1 + xn      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
6      1 + nx      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
7
8  #include <stdio.h>
9
10 int main()
11 {
12     int x, n, i;
13     float y, power = 1;
14
15     printf("Enter the value of x: ");
16     scanf("%d", &x);
17
18     printf("Enter the value of n: ");
```

Problems Output Debug Console Terminal Ports

- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_24.c -o Program_24 } ; if (\$?) { .\Program_24 }
Enter the value of x: 5
Enter the value of n: 6
Value of Y = 31.00
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_24.c -o Program_24 } ; if (\$?) { .\Program_24 }
Enter the value of x: 8
Enter the value of n: 2
Value of Y = 5.00
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_24.c -o Program_24 } ; if (\$?) { .\Program_24 }
Enter the value of x: 98
Enter the value of n: 56
Value of Y = 5489.00
- PS C:\CCC lab solutions> █

C Program_25.c X

C Program_25.c > main()

```
1 // PROGRAM TO PRINT TABLE OF ANY NO.
2
3 #include <stdio.h>
4
5 int main()
6 {
7     int n,i;
8
9     printf("Enter your number: ");
10    scanf("%d", &n);
11
12    for (i= 1 ; i <= 10; i++)
13    {
14        printf("%d x %d = %d\n", n,i,n*i);
15    }
16
17    return 0;
18 }
```

Problems Output Debug Console Terminal Ports

PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_25.c -o Program_25 } ; if (\$?) { .\Program_25 }

Enter your number: 11

```
11 x 1 = 11
11 x 2 = 22
11 x 3 = 33
11 x 4 = 44
11 x 5 = 55
11 x 6 = 66
11 x 7 = 77
11 x 8 = 88
11 x 9 = 99
11 x 10 = 110
```

PS C:\CCC lab solutions>

C Program_21.c C Program_26.c X

C Program_26.c > ...

```
1 //PROGRAM TO PRINT SUM OF ALL ODD AND EVEN NO B/W 1 TO N.
2
3 #include <stdio.h>
4
5 int main()
6 {
7     int n, i, oddSum = 0, evenSum = 0;
8
9     printf("Enter the value of N: ");
10    scanf("%d", &n);
11
12    for(i = 1; i <= n; i++)
13    {
14        if(i % 2 == 0)
15            evenSum = evenSum + i;
16        else
17            oddSum = oddSum + i;
18    }
```

Problems Output Debug Console Terminal Ports

- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_26.c -o Program_26 } ; if (\$?) { .\Program_26 }
Enter the value of N: 9
Sum of even numbers = 20
Sum of odd numbers = 25
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_26.c -o Program_26 } ; if (\$?) { .\Program_26 }
Enter the value of N: 8
Sum of even numbers = 20
Sum of odd numbers = 16
- PS C:\CCC lab solutions> cd "c:\CCC lab solutions\" ; if (\$?) { gcc Program_26.c -o Program_26 } ; if (\$?) { .\Program_26 }
Enter the value of N: 14
Sum of even numbers = 56
Sum of odd numbers = 49
- PS C:\CCC lab solutions> █